Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 36)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
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TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Describe the ecological features of mangroves and the adaptations of mangrove trees.

[3 marks]

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0 1 . 2 Explain **four** reasons why mangroves are important.

[4 marks]

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0 1 . 3 Describe the main threats to mangroves and the conservation efforts used.

[3 marks]

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0 2 Table 1 compares nitrogen fixation by Rhizobium bacteria in legume root nodules with the Haber process.

Table 1

Feature	Biological fixation	Haber process
Energy source	Sunlight captured by the plant (carbohydrates)	Fossil fuels (natural gas / coal)
Temperature	Soil temperature (10-30 °C)	About 450 °C
Pressure	Atmospheric	200 atmospheres
Catalyst	Nitrogenase enzyme	Iron
Energy density of input	Low	High
Waste products	None - all products used	CO_2, NOx

0 2 . 1 Use Table 1 to explain how natural processes differ from human industrial processes in their use of energy.

[4 marks]

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0 2 . 2 Give **three** other natural processes driven by solar energy that human society depends on.

[3 marks]

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0 2 . 3 Explain how the principle of low-temperature natural processes is being applied in human systems.

[3 marks]

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0 3 Table 1 shows some of the IUCN criteria used to place species in Red List categories.

Table 1

Category	Population decline over 10 years or 3 generations	Area of habitat	Number of mature adults	Probability of extinction in the wild
Vulnerable	over 40%	under 2000 km^2	under 10 000	over 10% in 100 years
Endangered	over 60%	under 500 km^2	under 2500	over 20% in 20 years or 5 generations
Critically endangered	over 80%	under 10 km^2	under 250	over 50% in 10 years or 3 generations

0 3 . 1 State **three** roles of the IUCN.

[3 marks]

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0 3 . 2 A species of lemur has declined from 12 000 to 2 100 mature adults in three generations and occupies 400 km^2 of forest. Use Table 1 to determine its Red List category. Explain your answer.

[3 marks]

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0 3 . 3 Explain why only a small proportion of species have been assessed for the Red List, and give **two** criteria used to decide which species to assess.

[3 marks]

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0 3 . 4 Give **one** example of a species whose Red List category improved because of conservation and state the actions responsible.

[1 mark]

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0 4 Table 1 shows typical energy values for an agroecosystem. Values are units of energy remaining from 100 000 units of solar energy.

Table 1

Stage	Energy / arbitrary units
Solar energy reaching the crop	100 000
Energy captured during photosynthesis	4 000
Energy stored in crop biomass	1 000

0 4 . 1 Calculate the percentage of the solar energy reaching the crop that is stored in crop biomass.

[1 mark]

Answer ____ %

0 4 . 2 Explain why so little of the solar energy is stored in the crop.

[3 marks]

0 4 . 3 Explain why few food chains have more than four trophic levels.

[2 marks]

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0 4 . 4 Explain why rearing sheep or cattle is the best option for food production in upland and semi-arid areas.

[2 marks]

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0 4 . 5 Explain how pigs can increase the efficiency of a food production system.

[2 marks]

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0 5 Table 1 shows some events in the history of life on Earth.

Table 1

Event	Time before present / billion years
Formation of the Earth	4.6
First life (Archaea) on Earth	3.5
First photosynthetic organisms	2.7
Oxygen concentration in the atmosphere starts to rise	2.45
End of the period of oxygen build-up (Proterozoic)	0.54

0 5 . 1 Use Table 1 to calculate how many **million** years passed between the appearance of photosynthetic organisms and the start of the rise in atmospheric oxygen.

[1 mark]

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Answer [ ] million years

0 5 . 2 Explain why atmospheric oxygen did not start to rise as soon as photosynthesis began.

[2 marks]

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0 5 . 3 Explain how the rise in atmospheric oxygen made life on the land surface possible.

[3 marks]

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0 5 . 4 Explain how early photosynthetic organisms helped to prevent a long-term rise in the Earth's temperature.

[2 marks]

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0 5 . 5 Give **two** other ways in which living organisms changed the conditions on Earth.

[2 marks]

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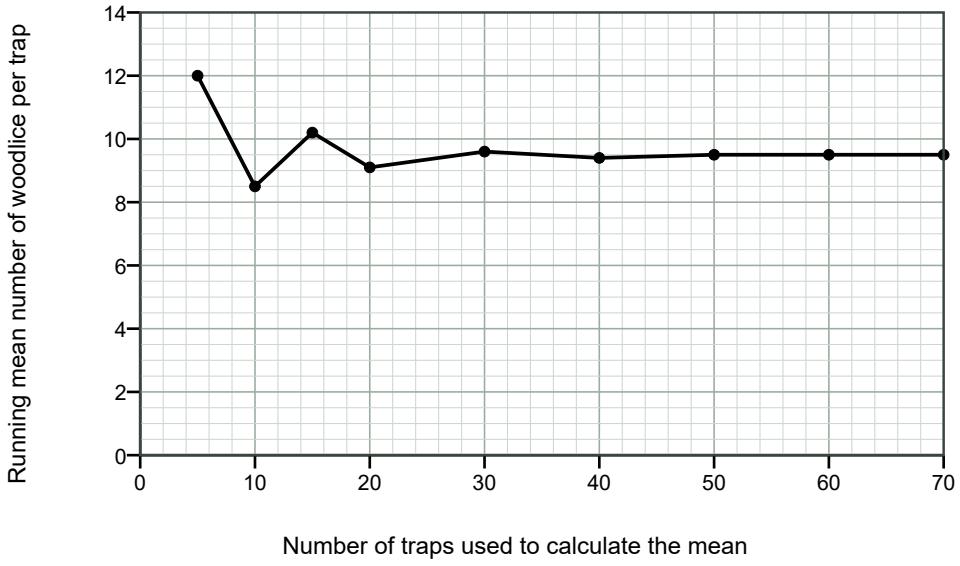
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0 6 Figure 1 shows the mean number of woodlice per pitfall trap calculated as more traps were included in the mean (a running mean).

Figure 1



0 6 . 1 Use Figure 1 to suggest the minimum number of traps needed to produce a reliable mean. Give a reason for your answer.

[2 marks]

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0 6 . 2 Explain why the number of samples needed depends on the degree of scatter of the results.

[2 marks]

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0 6 . 3 The student wants to compare woodlouse numbers in a woodland and in a grassland. Describe how the sampling should be standardised.

[3 marks]

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0 6 . 4 Explain why the timing of sampling would matter if the student were investigating the activity of flying insects rather than woodlice.

[3 marks]

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0 7 . 1 Explain why an electronic meter used to measure dissolved oxygen must be calibrated, and what should be done if the reading cannot be adjusted.

[2 marks]

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0 7 . 2 Give **three** types of indirect evidence of the presence of an animal and the information each can provide.

[3 marks]

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0 7 . 3 Explain how auditory monitoring can be used to study bats.

[2 marks]

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0 7 . 4 Explain **three** assumptions of the mark-release-recapture method that may not be met.

[3 marks]

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0 8 The Hawaiian goose (nene) was reduced by hunting, habitat loss and introduced predators to about 30 birds. A captive breeding programme started with just seven individuals and has increased the wild population to over 2000.

0 8 . 1 Explain why starting the programme with seven individuals caused problems, and describe how these problems are managed.

[4 marks]

0 8 . 2 Explain **three** reasons, other than gene pool size, why species kept in captivity may fail to breed.

[3 marks]

0 8 . 3 Describe how artificial insemination and embryo transfer increase breeding success in captive mammals.

[3 marks]

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0 9 . 1 Give **two** habitats created by farming that support important wildlife communities.

[2 marks]

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0 9 . 2 Explain how drainage and nutrient enrichment of farmland reduce biodiversity.

[2 marks]

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0 9 . 3 Explain **one** problem that can be caused by introducing a biological control species.

[1 mark]

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1 0 . 1 Evaluate the extent to which aquaculture can replace fishing as a source of aquatic food.

[9 marks]

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1 0 . 2 State what is meant by a **fed** aquaculture species.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably.

[25 marks]

OR

1 1 . 2 Discuss the extent to which agriculture can be made sustainable while feeding a growing human population.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS